Think — Project 3: fraction averages

A Cambridge project: the fraction that lies between two others, and where it always lands.

LESSON 133 | 1 × 40 MIN | MATEMATIKA 6, O'YLAB KO'R | S7 PROJECT 3

LESSON CLOCK

8' | 12' | 14' | 6'

The mean of two fractions	8 min
A strange rule	12 min
Testing the conjecture	14 min
Presenting it	6 min

LEARNING OBJECTIVES

- Find the mean of two fractions.
- Investigate what happens when numerators and denominators are added instead.
- State a conjecture and test it.
- Present the investigation clearly.

TEXTBOOK

National	pp. 389–391
Cambridge	Stage 7, project pages
Workbook	Project sheet 3

01 Explanation

The mean of two fractions

$$\text{mean} = \frac{a + b}{2}$$

TWO FRACTIONS	SUM	MEAN
$\frac{1}{2}, \frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{8}$
$\frac{1}{3}, \frac{1}{2}$	$\frac{5}{6}$	$\frac{5}{12}$
$\frac{2}{5}, \frac{3}{5}$	1	$\frac{1}{2}$

THE MEAN ALWAYS LIES BETWEEN THE TWO

That is the point of an average, and it is the check to make on every answer in this investigation.

A strange rule

Now try something that looks wrong: add the numerators and add the denominators.

$$\frac{a}{b} \oplus \frac{c}{d} = \frac{a+c}{b+d}$$

TWO FRACTIONS	THE STRANGE RULE	THE TRUE MEAN	BETWEEN THEM?
$\frac{1}{2}, \frac{1}{4}$	$\frac{2}{6} = \frac{1}{3}$	$\frac{3}{8}$	yes
$\frac{1}{3}, \frac{1}{2}$	$\frac{2}{5}$	$\frac{5}{12}$	yes
$\frac{1}{4}, \frac{3}{4}$	$\frac{4}{8} = \frac{1}{2}$	$\frac{1}{2}$	yes

THIS IS NOT HOW FRACTIONS ARE ADDED

$\frac{1}{2} + \frac{1}{4}$ is $\frac{3}{4}$, not $\frac{2}{6}$. The rule above is a different operation, worth investigating precisely because it is the mistake everyone makes.

Testing the conjecture

Conjecture: the result of the strange rule always lies between the two fractions.

TEST	RESULT	BETWEEN?
$\frac{1}{5}, \frac{4}{5}$	$\frac{5}{10} = \frac{1}{2}$	yes
$\frac{2}{3}, \frac{5}{7}$	$\frac{7}{10}$	yes — $0.667 < 0.7 < 0.714$
$\frac{1}{2}, \frac{1}{2}$	$\frac{2}{4} = \frac{1}{2}$	equal to both

THE CONJECTURE SURVIVES EVERY TEST

It is true, and it has a name: the **mediant** of two fractions. It always lies between them — but it is not usually the mean, as the first table showed.

Presenting it

PART	WHAT IT CONTAINS
the question	where does $\frac{a+c}{b+d}$ lie?
the data	at least six pairs, chosen systematically
the pattern	always between, but not the mean
the comparison	the mean against the mediant, in decimals
the conclusion	two sentences, with one worked example

WHERE THE MEDIANT IS USED

Two batsmen scoring *40* runs off *50* balls and *30* off *40* have a combined rate of $\frac{70}{90}$ — the mediant, not the mean of their two rates. Adding the tops and the bottoms is exactly right for combining rates.

02 Worked examples

EXAMPLE 1 **Find the mean of $\frac{1}{2}$ and $\frac{1}{4}$.**

1. $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$
2. Divide by *2*.
3. $\frac{3}{8}$

   Between $\frac{1}{4}$ and $\frac{1}{2}$ ✓

ANSWER

$\frac{3}{8}$

EXAMPLE 2 **Apply the strange rule to $\frac{1}{3}$ and $\frac{1}{2}$, and compare with the mean.**

1. Strange rule: $\frac{2}{5} = 0.4$.
2. Mean: $\frac{5}{12} \approx 0.417$.
3. Both lie between 0.333 and 0.5, but they differ.

ANSWER

$\frac{2}{5}$ against $\frac{5}{12}$

EXAMPLE 3 **A batsman scores *40* off *50* balls and then *30* off *40*. Find the combined rate.**

1. Total runs: 70. Total balls: 90.
2. $\frac{70}{90} \approx 0.78$ runs a ball.
3. That is the mediant of $\frac{40}{50}$ and $\frac{30}{40}$.

   Adding tops and bottoms is right here.

ANSWER

$\frac{7}{9}$

03 Interactive model

Point out that the “wrong” way to add fractions is exactly the right way to combine rates; the class remembers both rules better for seeing where each belongs.

Mean or mediant?

The mean of $\frac{1}{2}$ and $\frac{1}{4}$ is:

- $\frac{1}{3}$
- $\frac{3}{8}$
- $\frac{2}{6}$
- $\frac{3}{4}$

The mediant of the same two is:

- $\frac{1}{3}$
- $\frac{3}{8}$
- $\frac{3}{4}$
- $\frac{1}{6}$

$\frac{1}{2} + \frac{1}{4}$ equals:

- $\frac{2}{6}$
- $\frac{3}{4}$
- $\frac{1}{3}$
- $\frac{1}{8}$

The mediant always lies:

- above both
- below both
- between them
- anywhere

Is the mediant the mean?

- always
- sometimes
- never
- only for equal fractions

Two rates $\frac{40}{50}$ and $\frac{30}{40}$ combine to:

- $\frac{70}{90}$
- $\frac{70}{45}$
- $\frac{35}{45}$
- $\frac{1200}{2000}$

Adding tops and bottoms is right for:

- adding fractions
- combining rates
- both
- neither

A conjecture becomes a fact when it is:

- tested twice
- proved
- written down
- believed

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Mean	O‘rta arifmetik	Среднее арифметическое
Between	Orasida	Между
Conjecture	Faraz	Гипотеза
To test	Sinab ko‘rmoq	Проверить
Numerator	Surat	Числитель
Denominator	Maxraj	Знаменатель
Investigation	Tadqiqot	Исследование
Counter-example	Qarshi misol	Контрпример

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The mean of two numbers is:

- their sum
- half their sum
- their difference
- their product

The mediant of $\frac{a}{b}$ and $\frac{c}{d}$ is:

- $\frac{a + c}{b + d}$
- $\frac{a + c}{2}$
- $\frac{ac}{bd}$
- $\frac{a}{b} + \frac{c}{d}$

The mediant of $\frac{1}{4}$ and $\frac{3}{4}$ is:

- $\frac{1}{2}$
- $\frac{4}{4}$
- $\frac{3}{16}$
- $\frac{1}{8}$

Is that also the mean here?

- yes
- no
- sometimes
- never

Adding fractions needs:

- a common denominator
- adding the denominators
- the mediant
- nothing

The strange rule is right for:

- fractions
- rates
- both
- neither

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. The mean of $\frac{1}{2}$ and $\frac{1}{4}$

   ANSWER $\frac{3}{8}$

2. The mean of $\frac{1}{3}$ and $\frac{1}{2}$

   ANSWER $\frac{5}{12}$

3. The mean of $\frac{2}{5}$ and $\frac{3}{5}$

   ANSWER $\frac{1}{2}$

4. The mediant of $\frac{1}{2}$ and $\frac{1}{4}$

   ANSWER $\frac{1}{3}$

5. The mediant of $\frac{1}{3}$ and $\frac{1}{2}$

   ANSWER $\frac{2}{5}$

6. The mediant of $\frac{1}{4}$ and $\frac{3}{4}$

   ANSWER $\frac{1}{2}$

7. $\frac{1}{2} + \frac{1}{4}$

   ANSWER $\frac{3}{4}$

Medium · the standard question

7 PROBLEMS

1. Is the mediant always between the two fractions?

   ANSWER Yes

2. Is the mediant always the mean?

   ANSWER No

3. When are they equal?

   ANSWER When the two fractions are equal, or the denominators match

4. The mediant of $\frac{1}{5}$ and $\frac{4}{5}$

   ANSWER $\frac{1}{2}$

5. The mediant of $\frac{2}{3}$ and $\frac{5}{7}$

   ANSWER $\frac{7}{10}$

6. Combine 40 off 50 and 30 off 40

   ANSWER $\frac{7}{9}$

7. Which rule adds fractions?

   ANSWER The common denominator

Hard · stretch and reason

7 PROBLEMS

1. Show that $\frac{7}{10}$ lies between $\frac{2}{3}$ and $\frac{5}{7}$

   ANSWER $0.667 < 0.7 < 0.714$

2. The mean of $\frac{2}{3}$ and $\frac{5}{7}$

   ANSWER $\frac{29}{42}$

3. Which is larger, the mean or the mediant, for those two?

   ANSWER The mean, at 0.690 against 0.7 — the mediant

4. A pupil scores $18/25$ and $27/35$: the combined mark

   ANSWER $\frac{45}{60} = 75\%$

5. Why is the mediant right for combining marks?

   ANSWER Totals over totals is what a combined mark means

6. Find a fraction between $\frac{3}{7}$ and $\frac{4}{7}$

   ANSWER $\frac{7}{14} = \frac{1}{2}$

7. Find a fraction between $\frac{5}{8}$ and $\frac{2}{3}$

   ANSWER $\frac{7}{11}$

07 Homework

Homework — the project

One page: six tested pairs, the pattern, and where the rule is genuinely useful.

1. Find the mean and the mediant of six pairs of fractions of your own.

2. Record both in a table, with decimals for comparison.
3. State in one sentence where the mediant always lies.
4. Give one example where adding tops and bottoms is the right thing to do.
5. Explain why it is nevertheless the wrong way to add two fractions.